INTERNAL EXCELLENCE, EXTERNAL COLLAPSE: A RIGOROUS AUDIT OF A KNOWLEDGE-DISTILLED CERVICAL CYTOLOGY CLASSIFIER

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ABSTRACT

We audit a two-stage EfficientNet-B4 teacher knowledge-distilled into an EfficientNet-B0 student for five-class cervical-cell classification on **SIPaKMeD**. In-distribution the distilled student is excellent—96.6% accuracy, 0.967 macro-F_1, 0.998 one-vs-rest AUROC, and an expected calibration error (ECE) of only 0.015—and it surpasses its own teacher (94.3%). Grad-CAM confirms the model attends to nuclear/cellular structure. Yet under a binary normal/abnormal collapse the same model *collapses out-of-distribution*: 63.9% on Herlev, 60.5% on HiCervix and 53.2% on APACC, with AUROCs of 0.73, 0.59 and 0.53. We diagnose the cause in detail and show it is *not* a labelling artefact: the predicted $P(\text{abnormal})$ ranks abnormal above normal cells in every dataset (positive separation, AUROC>0.5). Instead the failure is (i) genuine domain-shift signal degradation that tracks slide preparation similarity (conventional smear $\rightarrow$ liquid-based $\rightarrow$ auto-segmented crops), and (ii) a per-domain shift of the decision score that makes the fixed τ=0.5 operating point underperform even a majority-class baseline despite real ranking signal. The internal number is itself optimistic because it is measured on a leaky random split. We conclude that headline accuracy is a poor proxy for screening readiness, and that the remedies are leakage-safe evaluation, stain/domain normalization and per-domain calibration—not relabelling.

Keywords cervical cytology · knowledge distillation · domain shift · calibration · SIPaKMeD

1 Introduction

Automated cervical-cytology screening is dominated by lightweight, edge-deployable models trained and evaluated *within a single dataset*, where reported accuracies routinely exceed 97%. Our starting point is exactly such a model: an EfficientNet-B4 teacher [Tan and Le, 2019] distilled [Hinton et al., 2014] into an EfficientNet-B0 student on SIPaKMeD [Plissiti et al., 2018]. This report asks a single question—*does it generalize?*—and answers it with an honest cross-dataset audit. We report the full in-distribution performance, calibration and Grad-CAM explanations, then the cross-dataset results, and finally a detailed diagnosis of why the model fails to transfer.

2 Methods

Data. SIPaKMeD provides $4{,}049$ cropped single cells in five cytomorphology classes (Superficial-Intermediate, Parabasal, Metaplastic, Koilocytotic, Dyskeratotic). **Model.** The B4 teacher is trained in two stages (frozen backbone, then fine-tuned), and a B0 student is trained by distillation with the canonical temperature-scaled logit loss $\mathcal{L} = \alpha\,\mathrm{CE} + (1-\alpha)\,T^2\,\mathrm{KL}(\sigma(z_t/T)\,\|\,\sigma(z_s/T))$. **Cross-dataset protocol.** The native label spaces of the external sets do not match SIPaKMeD's five classes, so both the prediction and the ground truth are collapsed to a binary normal/abnormal decision, scoring $P(\text{abnormal}) = \sigma(z)_{\text{Koilo}} + \sigma(z)_{\text{Dysk}}$ against a pre-registered threshold τ=0.5. Glandular and indeterminate native classes have no home in the squamous five-class space and are excluded (and the excluded fraction reported).

3 In-distribution results

Training. The teacher converges over 71 epochs (30 frozen + 41 fine-tuning, early-stopped) and the student over 34 epochs; Figure 1 shows smooth, non-overfit train/validation curves.

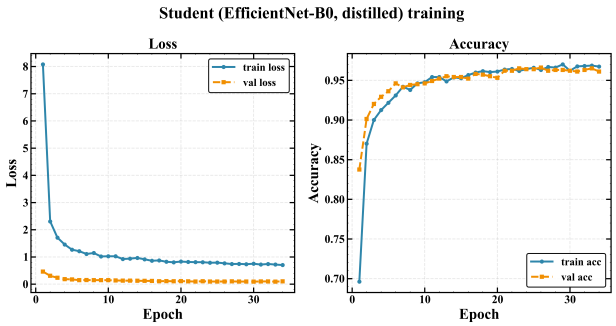


Figure 1: Distilled B0 student training: train/validation loss and accuracy. Validation tracks training closely (no overfitting), converging to $\approx 96\%$ validation accuracy.

Table 1: In-distribution five-class metrics (SIPaKMeD validation). The distilled student surpasses the teacher on all metrics and is well calibrated.

Metric	Teacher (B4)	**Student (B0)**
Accuracy	0.943	**0.966**
Balanced accuracy	0.946	**0.968**
Macro-F_1	0.945	**0.967**
Cohen's κ	0.929	**0.958**
AUROC (OvR)	0.996	**0.998**
ECE	0.013	0.015

Classification. Table 1 reports the five-class metrics. The distilled student reaches 96.6% accuracy and *exceeds* its teacher on every metric—evidence that the distillation signal is informative rather than inert. Per-class F_1 (student) is highest for Superficial-Intermediate (0.995) and Parabasal (0.994) and lowest for Koilocytotic (0.919, recall 0.873), the class most often confused with its neighbours (Figure 2).

Calibration. Both models are well calibrated: ECE $=$ 0.013 (teacher) and 0.015 (student). The reliability diagram (Figure 3) lies close to the identity, meaning the in-distribution confidences are trustworthy—a property we show *does not* survive the domain shift.

4 Explainability

Grad-CAM [Selvaraju et al., 2017] over the student's final convolutional block (Figure 4) localizes the evidence to the nucleus and cell body for one representative correctly-classified cell per class (confidences 0.97–1.00). The attributions are cytologically plausible, indicating the in-distribution decisions are driven by cell structure rather than background artefacts.

5 Cross-dataset generalization

We evaluate the SIPaKMeD-trained student, unchanged, on three external single-cell Pap datasets at the binary level. Table 2 and Figure 5 report the result: accuracy falls from 96.6% internally to 53–64% externally, with AUROCs of 0.53–0.73. Sensitivity on HiCervix is only 0.39—the model *misses most abnormal cells* out of distribution.

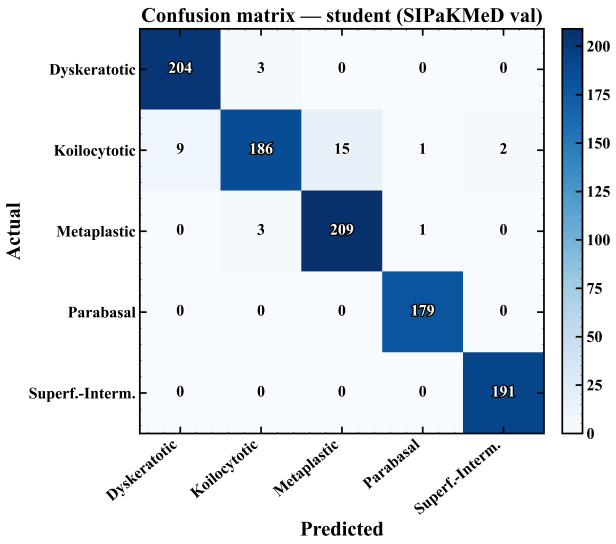


Figure 2: Five-class confusion matrix (student). Errors concentrate on Koilocytotic, which is confused with the other abnormal/metaplastic classes.

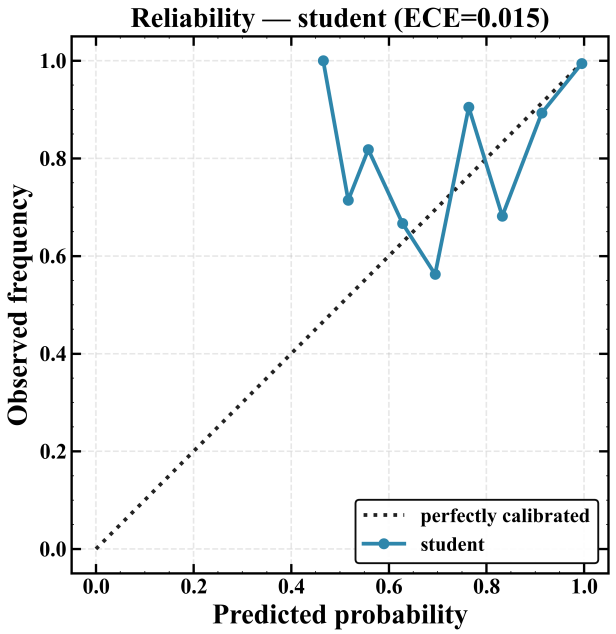


Figure 3: Reliability diagram (student, ECE $= 0.015$). In-distribution confidence is well calibrated; bins lie close to the diagonal.

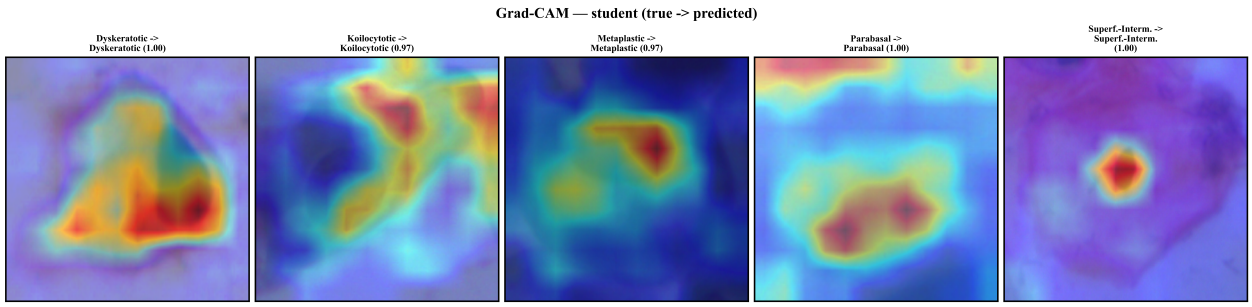


Figure 4: Grad-CAM for the student, one representative cell per SIPaKMeD class (true → predicted, confidence). Heatmaps concentrate on nuclear/cellular regions.

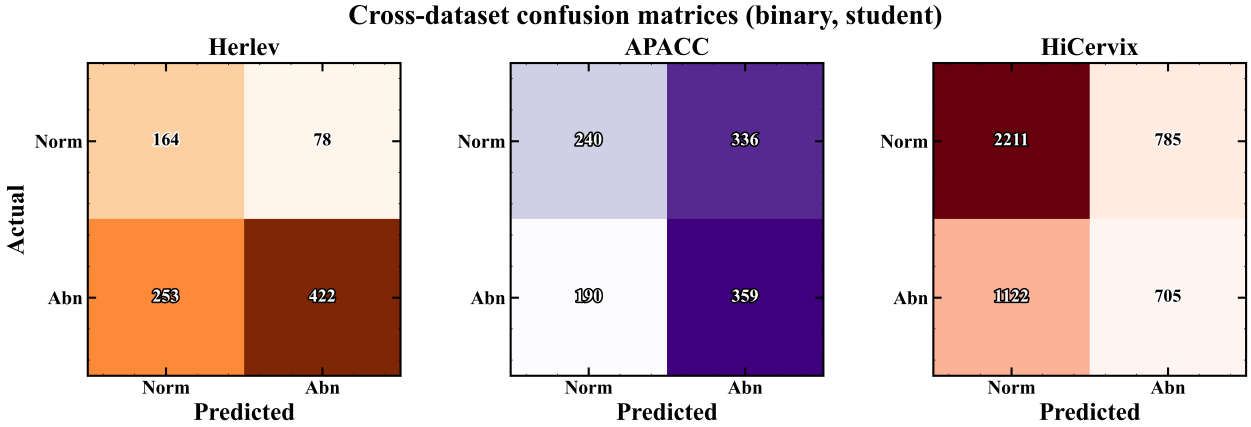


Figure 5: Cross-dataset binary confusion matrices (student). The model over-predicts *abnormal* on APACC (low specificity) and *normal* on HiCervix (low sensitivity).

Table 2: Cross-dataset binary (normal/abnormal) results for the student. ΔP is the separation between mean $P(\text{abnormal})$ on true-abnormal vs. true-normal cells (positive = correct ordering). Internal SIPaKMeD accuracy is 0.966.

Dataset	n	Acc	Sens	Spec	AUROC
Herlev	917	0.639	0.625	0.678	**0.728**
HiCervix	4823	0.605	0.386	0.738	0.594
APACC	1125	0.532	0.654	0.417	0.525

6 Why cross-dataset evaluation fails

(1) It is not a labelling artefact. A flipped or mis-defined mapping would make the decision score rank abnormal *below* normal cells, giving AUROC< 0.5. The opposite holds: the separation ΔP between mean $P(\text{abnormal})$ on true-abnormal and true-normal cells is *positive everywhere*—$+0.284$ (Herlev), $+0.112$ (HiCervix), $+0.045$ (APACC)—and every AUROC exceeds 0.5 (Figure 6). The per-dataset mappers were audited explicitly (Herlev 7 →binary, APACC healthy/unhealthy, HiCervix Bethesda `level_1`: negative→normal, ASC→abnormal, AGC/microbe→excluded). The labels are correctly oriented; the model simply has little to rank with.

(2) Operating-point miscalibration. The well-calibrated in-distribution confidence does not transfer: the $P(\text{abnormal})$ distribution shifts per domain (baseline ≈ 0.6 on APACC → over-calls abnormal; ≈ 0.35 on HiCervix → under-calls). At the fixed τ=0.5 the binary accuracy is therefore *below the majority-class baseline* for both Herlev ($0.639 < 0.74$) and HiCervix ($0.605 < 0.62$), even though the AUROC shows real ranking signal. A per-domain optimal threshold recovers much of it (Herlev 0.75 at τ=0.05; HiCervix 0.65 at τ=0.85), but choosing τ on the test set is not an honest operating point—it only quantifies the calibration gap.

(3) The signal degrades with preparation distance. The retained AUROC tracks how similar each dataset's slide preparation is to SIPaKMeD's conventional smear: Herlev (conventional smear) $0.73 >$ HiCervix (liquid-based) $0.59 >$ APACC (auto-segmented, noisy crops) $0.53 \approx$ chance. The collapse is dominated by preparation/scanner/segmentation shift, exactly the axis a single-source model is blind to.

(4) The internal number is optimistic. The 96.6% is measured on a random per-cell validation split, which

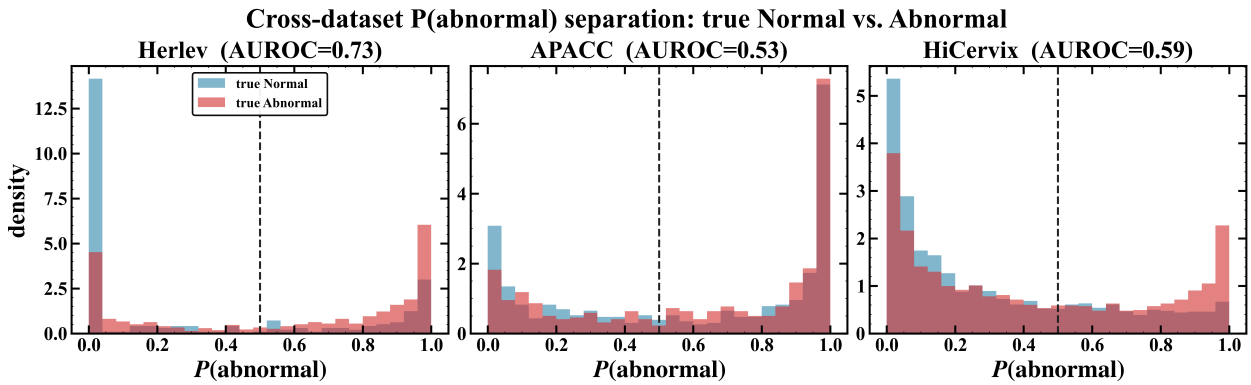


Figure 6: Distribution of the decision score $P(\text{abnormal})$ for true-normal (blue) vs. true-abnormal (red) cells. Separation is clear on Herlev (AUROC 0.73), weak on HiCervix (0.59) and essentially absent on APACC (0.53). The dashed line is the τ=0.5 threshold.

leaks same field-of-view (FoV) cells across train and validation; a leakage-safe, cluster-grouped split would lower it. The true internal→external gap is therefore even larger than the ≈ 33-point drop observed here.

7 Conclusion

The audited model is internally excellent (96.6%, well calibrated, cytologically faithful Grad-CAM) yet *does not generalize*: binary AUROC falls to 0.53–0.73 across datasets and binary accuracy drops below trivial baselines under a fixed threshold. Crucially, this is not a labelling error—the decision score is correctly oriented in every dataset—but a combination of domain-shift signal loss and operating-point miscalibration, on top of an optimistic leaky internal estimate. Headline single-dataset accuracy is thus a poor proxy for screening readiness. The indicated remedies are leakage-safe (cluster-grouped) evaluation, stain/domain normalization [Macenko et al., 2009] and per-domain calibration [Guo et al., 2017], and reporting threshold-free AUROC alongside the excluded fraction—not relabelling.

References

Mingxing Tan and Quoc Le. EfficientNet: Rethinking model scaling for convolutional neural networks. In *International Conference on Machine Learning (ICML)*, pages 6105–6114, 2019.

Geoffrey Hinton, Oriol Vinyals, and Jeff Dean. Distilling the knowledge in a neural network. *NIPS Deep Learning and Representation Learning Workshop (arXiv:1503.02531)*, 2014.

Marina E. Plissiti, P. Dimitrakopoulos, G. Sfikas, C. Nikou, O. Krikoni, and A. Charchanti. SIPaKMeD: A new dataset for feature and image based classification of normal and pathological cervical cells in Pap smear images. In *IEEE International Conference on Image Processing (ICIP)*, pages 3144–3148, 2018.

Ramprasaath R. Selvaraju, Michael Cogswell, Abhishek Das, Ramakrishna Vedantam, Devi Parikh, and Dhruv Batra. Grad-CAM: Visual explanations from deep networks via gradient-based localization. In *IEEE International Conference on Computer Vision (ICCV)*, pages 618–626, 2017.

Marc Macenko, M. Niethammer, J.S. Marron, D. Borland, J.T. Woosley, X. Guan, C. Schmitt, and N.E. Thomas. A method for normalizing histology slides for quantitative analysis. In *IEEE International Symposium on Biomedical Imaging (ISBI)*, pages 1107–1110, 2009.

Chuan Guo, Geoff Pleiss, Yu Sun, and Kilian Q. Weinberger. On calibration of modern neural networks. In *International Conference on Machine Learning (ICML)*, pages 1321–1330, 2017.